

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

AU-5001-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Machine Component Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain the basic procedure for mechanical engineering design.
b) Discuss the role of an industrial designer in product design.
2. a) How the shaft is designed when it is subjected to twisting moment only.
b) A shaft supported at the ends in ball bearings carries a straight tooth spur gear at its mid span and is to transmit 7.5 kW at 300 r.p.m. The pitch circle diameter of the gear is 150mm. The distances between the centre line of bearings and gear are 100mm each. If the shaft is made of steel and the allowable shear stress is 45MPa, determine the diameter of the shaft. Show in a sketch how the gear will be mounted on the shaft; also indicate the ends where the bearings will be mounted? The pressure angle of the gear may be taken as 20°.

AU-5001-CBGS

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[2]

3. A propeller shaft is required to transmit 50kW power at 600rpm. It is a hollow shaft having an inside dia. 0.8 times of the outside diameter. It is made of steel and factor of safety is 4. Calculate the inside and outside dia.
4.
 - a) What is the difference between rigid and flexible coupling?
 - b) What are different types of threads for power screw?
5.
 - a) What are the materials used for lining of friction surfaces of a clutch.
 - b) An automotive plate clutch consists of two pair of contacting surfaces with an asbestos friction lining. The torque transmitting capacity of the clutch is 550N-m. The coefficient of friction is 0.25 and the permissible intensity of pressure is 0.5N/mm^2 . Due to space limitations, the outer diameter of the friction disk is fixed as 250mm. Using the uniform wear theory, calculate:
 - i) The inner diameter of the friction disk
 - ii) The spring force required to keep the clutch in engaged position.
6. A multidisk clutch consists of five steel plates and four bronze plates and transmits 10kW power at 750rpm. The ratio of outer diameter to inner disks is 2. The coefficient of friction is 0.1 and the intensity of pressure on friction lining is limited to 0.3 N/mm^2 , Assume uniform wear theory, calculate the inner and outer diameters of friction disks.

[3]

7. a) Explain about the Lewis equation to determine the beam strength of gear teeth.
- b) A gear drive is required to transmit a maximum power of 22.5kW. The velocity ratio is 1:2 and r.p.m. of the pinion is 200. The approximate centre distance between the shafts may be taken as 600mm. The teeth has 20° stud involute profiles. The static stress for the gear material (which is cast iron) may be taken as 60MPa and face width as 10 times the module. Find the module, face width, and number of teeth on each gear.
8. Write short notes on any two
- i) Design for safety and cost.
 - ii) Steps in design of helical gear.
 - iii) Design of band brake.
